MODULE 56 · DOMAIN 4 - RESILIENCE, MIGRATION AND CONTINUITY

Practice Exam: Domain 4

Fifteen original questions on resilience, migration and business continuity. Pause, decide, then watch.

DOMAIN 4 of 5 EXAM WEIGHT 15% SET 4 of 6

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How to use this set

- **Allow 2.4 minutes per question**: 36 minutes for the set.
- **Write your answer down** before you unpause.
- **Do the arithmetic**: RTO, RPO, terabytes and bandwidth.
- Watch for services closed to new customers, such as Migration Hub.

> ON THE EXAM
>
> Numbers decide resilience and migration questions. Check them before weighing the options.

> A defence supplier is planning a migration from a site with no internet access. No software agents may be installed on its servers.
>
> It needs server inventory, utilisation and dependency data to plan waves.

What should the solutions architect recommend?

A. Install Application Discovery Service agents on each server

B. Use AWS Migration Hub network visualisation

C. Deploy the AWS Transform discovery appliance on premises, then upload its exported data

D. Interview application owners and build a spreadsheet

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Discovery agents
Agents are prohibited; service closed to new customers.

B Migration Hub
Closed to new customers.

C Discovery appliance, export
Agentless, works offline, data uploaded later.

D Interviews
No utilisation or dependency data.

THE SIGNAL

No agents allowed, air-gapped site: deploy the discovery appliance on premises and upload its export. Module 32.

> During wave planning, a dependency map shows that an order application and a pricing service make thousands of synchronous calls per transaction to each other.
>
> The network link to AWS adds 15 milliseconds of latency.

How should the solutions architect plan the migration?

A Move both in the same wave

B Move the order application first, then pricing a month later

C Move pricing first to test the network

D Rewrite both before migrating

◆ PAUSE AND DECIDE

How should the solutions architect plan the migration?

A Same wave

Chatty dependencies stay on the same side.

B Order first

Thousands of calls cross the link each transaction.

C Pricing first

Same latency problem, reversed.

D Rewrite first

Not needed to migrate.

THE SIGNAL

Tightly coupled, chatty dependencies migrate together. Module 32.

A hospital is rehosting 400 VMware virtual machines to EC2. Its change board forbids installing any new software inside the guest operating systems.

Replication must be continuous so that cutover takes minutes.

What should the solutions architect use?

A AWS Transform MGN with the replication agent on each VM

B VM Import/Export of nightly snapshots

C Rebuild each server from scratch on EC2

D AWS Transform MGN with agentless replication through the vCenter appliance

◆ PAUSE AND DECIDE

What should the solutions architect use?

A MGN with agents
Installs software in every guest.

B VM Import/Export
Not continuous; long cutovers.

C Rebuild
Months of work, not a rehost.

D MGN agentless
Continuous replication with nothing in the guests.

THE SIGNAL

No software allowed inside VMware guests: agentless replication through the vCenter appliance. Module 33.

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> A retailer must leave its data centre within eight weeks because the lease ends. It runs 900 VMs on VMware Cloud Foundation and holds VCF licences.
>
> Operations staff must keep their existing VMware tools and processes for now.

What should the solutions architect recommend?

A Rehost all 900 VMs to EC2 with AWS Transform MGN

B Relocate to Amazon Elastic VMware Service

C Refactor the applications to containers

D Replatform onto Amazon RDS and Fargate

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Rehost to EC2

Changes operations, too much in eight weeks.

B Relocate to EVS

Same VMware stack and tools, own licences, fast.

C Refactor

Far too long.

D Replatform

Far too long, and changes operations.

THE SIGNAL

Exit fast, change nothing in operations, keep the VCF licences: relocate to Amazon EVS. Module 33.

> A company is moving a 6 TB Oracle database to Aurora PostgreSQL. It uses PL/SQL stored procedures heavily.
>
> Downtime at cutover must be under 15 minutes.

What should the solutions architect do?

A Export with Data Pump and import into Aurora

B Use AWS DMS full load only during a weekend outage

C Use Aurora zero-ETL integration

D Convert the schema and code with DMS Schema Conversion, then use DMS full load plus ongoing replication and cut over

◆ PAUSE AND DECIDE

What should the solutions architect do?

A Data Pump
Oracle-only tooling; long outage.

B Full load only
Hours of downtime for 6 TB.

C Zero-ETL
Feeds analytics, not a migration.

D Schema Conversion, full load plus CDC
Converts code, then keeps data in sync until cutover.

THE SIGNAL

Changing engine adds a conversion step before any data moves. Module 34.

A research institute must copy 500 TB from on-premises NFS servers to S3 within two weeks. It has an unused 10 Gbps Direct Connect connection.

Transfers must be verified and must not saturate the link during working hours.

What should the solutions architect use?

A Order Snowball Edge devices

B A Python script using the S3 API

C AWS DataSync over Direct Connect, with a bandwidth limit schedule and verification

D S3 File Gateway

◆ PAUSE AND DECIDE

What should the solutions architect use?

A Snowball Edge

Closed to new customers, and slower here.

B Custom script

No verification or throttling built in.

C DataSync

Fits in about six days at line rate; throttles and verifies.

D File Gateway

For ongoing file access, not a bulk copy.

THE SIGNAL

Bulk online transfer with verification and scheduling: DataSync. Module 34.

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A company runs MySQL on self-managed EC2 instances. The DBA team spends days each month on patching, backups and failover testing.

The application team cannot change more than connection strings.

Which migration strategy should the solutions architect recommend?

A Refactor the application to DynamoDB

B Replatform to Amazon Aurora MySQL

C Rehost onto larger EC2 instances

D Retire the database

◆ PAUSE AND DECIDE

Which migration strategy should the solutions architect recommend?

A Refactor to DynamoDB
Major code changes.

B Replatform to Aurora
Managed patching, backups, failover; same engine.

C Rehost bigger
Same operational work.

D Retire
The data is needed.

THE SIGNAL

Reduce operations with little code change: replatform. New capabilities: refactor. Module 35.

A bank's core system runs as 4 million lines of COBOL on a mainframe. It wants to keep the business logic exactly as it behaves today, but run it as Java on AWS.

The bank wants to automate as much of the conversion as possible.

What should the solutions architect recommend?

A AWS Transform for mainframe, using its refactor path to convert COBOL to Java

B Rewrite the system from scratch as microservices

C Rehost the mainframe on EC2 with an emulator

D The Mainframe Modernization managed runtime environment

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Transform, refactor to Java
Automated conversion, same logic, new language.

B Rewrite
Changes the design; years of work.

C Emulator rehost
Keeps COBOL.

D Managed runtime
Closed to new customers.

THE SIGNAL

Keep how it works, change the language: refactor. Keep what it decides, change how: reimagine. Module 36.

A logistics firm wants disaster recovery for 300 servers in another Region. It needs an RPO of seconds and an RTO under 30 minutes.

While idle, the recovery site must cost as little as possible.

What should the solutions architect use?

A. Daily AWS Backup copies to the second Region

B. A full active-active deployment

C. AWS Elastic Disaster Recovery, replicating to a low-cost staging area

D. Warm standby with every server running at half size

◆ PAUSE AND DECIDE

What should the solutions architect use?

A Daily backups
RPO of a day.

B Active-active
Most expensive option.

C Elastic Disaster Recovery
Continuous replication, cheap staging, launch on failover.

D Warm standby
Pays for 300 running servers.

THE SIGNAL

Seconds of RPO, minutes of RTO, minimal idle cost, many servers: Elastic Disaster Recovery. Module 37.

A payments platform fails over between Regions by promoting an Aurora global database, scaling up ECS services and shifting Route 53 records. Today an engineer follows a 40-step document.

The company wants one orchestrated, approved plan that runs from the recovery Region, with progress visible to executives.

What should the solutions architect implement?

A. A Lambda function that runs the steps

B. Route 53 health checks with failover records

C. AWS Fault Injection Service

D. An Amazon Application Recovery Controller Region switch plan

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Lambda function
Custom, untested orchestration.

B Route 53 failover
Shifts DNS only.

C FIS
Tests failures; does not run recovery.

D ARC Region switch
Orchestrates database, scaling and traffic steps.

THE SIGNAL

Many recovery steps across services, one approved plan: ARC Region switch. Module 37.

A streaming company wants to prove its service survives losing a dependency, such as a database failover, during business hours in production.

Experiments must stop automatically if customer-facing errors exceed a threshold.

What should the solutions architect use?

A. A script that reboots random instances

B. AWS Fault Injection Service experiments with CloudWatch alarm stop conditions

C. AWS Resilience Hub assessments

D. A tabletop exercise

◆ PAUSE AND DECIDE

What should the solutions architect use?

A Random reboots

No targeting, no stop condition.

B FIS with stop conditions

Controlled faults, automatic stop on alarms.

C Resilience Hub

Assesses; does not inject faults.

D Tabletop

Does not test real behaviour.

THE SIGNAL

Inject faults in production safely: FIS with stop conditions. Module 38.

A bank must show that each of its 40 applications can meet its stated RTO and RPO. It must also catch changes that weaken resilience before they reach production.

It wants recommendations, not only a pass or fail.

What should the solutions architect use?

A AWS Resilience Hub, with resiliency policies per application and assessments run from the deployment pipeline

B AWS Fault Injection Service only

C A yearly disaster recovery test

D AWS Config conformance packs

◆ PAUSE AND DECIDE

What should the solutions architect use?

A Resilience Hub in the pipeline
Assesses against RTO and RPO, with recommendations.

B FIS only
Tests behaviour, not targets.

C Yearly test
Too infrequent to catch changes.

D Conformance packs
Configuration rules, not recovery objectives.

THE SIGNAL

Measure against RTO and RPO targets, continuously, with recommendations: Resilience Hub. Module 38.

> After a ransomware attack at a peer firm, a company wants backups that can be restored even if attackers gain administrator access to its production accounts.
>
> Recovery must not depend on any credentials in those accounts.

What should the solutions architect implement?

A S3 versioning on all buckets

B Daily snapshots in the same account

C An AWS Backup logically air-gapped vault, shared to a recovery account, with multi-party approval for access

D Copy backups to a second Region in the same account

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Versioning
Administrators can delete versions.

B Same-account snapshots
Deletable by the attacker.

C Logically air-gapped vault
Locked, separately owned, recoverable with approval.

D Second Region, same account
Same credentials reach both Regions.

THE SIGNAL

Recover even if the account is compromised: a logically air-gapped vault. Module 39.

A retailer runs all 60 microservices on one large EKS cluster. A failed control plane add-on upgrade made every service unavailable for two hours.

The company wants to limit the impact of future cluster problems.

What should the solutions architect recommend?

A Add more nodes to the cluster

B Split workloads across several clusters as separate failure domains, and upgrade them one at a time

C Stop upgrading the cluster

D Move the cluster to a larger instance type

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A More nodes

The cluster is still one failure domain.

B Several clusters

A cluster failure affects only its share.

C No upgrades

Unsupported versions and security risk.

D Larger instances

Same single failure domain.

THE SIGNAL

A cluster is itself a failure domain. Upgrades and misconfiguration can take a whole cluster down. Module 39.

> A stock exchange's member firms connect to its order API in two Regions. The firms allow-list specific IP addresses, and some of their systems cache DNS for hours.
>
> Regional failover must take effect within a minute.

What should the solutions architect use?

A Route 53 failover records with a 60-second TTL

B Route 53 latency-based records

C CloudFront in front of both Regions

D AWS Global Accelerator with endpoint groups in both Regions

◆ PAUSE AND DECIDE

What should the solutions architect use?

A Route 53 failover

Cached DNS ignores the TTL; IPs change.

B Latency records

Same DNS caching problem.

C CloudFront

Dynamic APIs; IP addresses change.

D Global Accelerator

Static IPs; shifts traffic without DNS changes.

THE SIGNAL

Clients that allow-list IPs or ignore DNS TTLs: Global Accelerator. Module 37.

Where to revisit

IF YOU MISSED	REWATCH
Q1, Q2	Module 32: Discovery and wave planning
Q3, Q4	Module 33: Rehost with Transform MGN and EVS
Q5, Q6	Module 34: Data migration
Q7	Module 35: Replatform and refactor
Q8	Module 36: Mainframe and legacy
Q9, Q10, Q15	Module 37: Disaster recovery design
Q11, Q12	Module 38: DR testing and chaos engineering
Q13, Q14	Module 39: Resilience and backup

ON THE EXAM

12 or more out of 15 is on track. If arithmetic caught you out, rework the numbers before moving on.